

RNA Processing Module: Convergence of Two Attention Mechanisms

RNA Self-Attention

Community 1 (679 genes)

RNA Binding
 $p = 5 \times 10^{-14}$

547
genes

DNA→RNA Cross-Attention

Community 3 (165 genes)

Spliceosome
 $p = 4 \times 10^{-9}$

132
core genes
(80%)

33
genes

Core 132 genes include:

- **Spliceosome (14):** SF3A3, SF3B4, PRPF3, SNRNP27, SNRPG, BCAS2, ...
- **RNA processing (14):** YTHDF2, ADAR, EXOSC10, TUT4, XPO1, ZFP36L2, ...
- **Chromatin (6):** HDAC1, KDM1A, ARID1A, RBBP4, ASH1L, RNF2
- **Transcription (5):** CDC73, TAF12, TRIM33, MEAF6, YY1AP1